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Multi-Label ICD-10 Classification of Greek Cardiology Discharge Summaries

System Description for the ElCardioCC 2026 Shared Task

Authors will be added later

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Abstract

We present our system for the ElCardioCC 2026 shared task on automatic ICD-10 coding of Greek cardiology discharge summaries. The system combines six heterogeneous components — a fine-tuned Greek BERT model, two XLM-RoBERTa variants (base and large), a hybrid information retrieval module, a dictionary-based rule system, and a named entity recognition with entity linking pipeline — under a declarative metaheuristic ensemble framework. Individual models are optimised with task-specific loss functions (Asymmetric Loss, ZLPR) and per-label threshold tuning on the validation set. The ensemble fuses component outputs through ten base strategies and four composition operators, with weights searched via random restarts, hill-climbing, and Variable Neighbourhood Search. Our best submission achieves a micro-F1 of **0.8667** on the official test set, improving by approximately 1.8 F1 points over the single best standalone model.

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1 Introduction

Clinical coding is the process of assigning standardised diagnostic codes to free-text medical records. It is a mandatory step in hospital administration, epidemiological surveillance, and insurance reimbursement. Manual coding is slow, expensive, and prone to inter-annotator disagreement, making automation an active research priority.

The ElCardioCC 2026 shared task targets Greek cardiology discharge summaries, requiring systems to predict all applicable codes from the WHO ICD-10 classification for each document. The setting is particularly challenging for three reasons:

1. **Low-resource morphologically rich language.** Greek has rich inflection and clinical Greek text mixes formal terminology with transliterated English abbreviations (e.g. *stent*, *TAVI*, *PCI*).
2. **Long documents.** Discharge summaries regularly exceed the 512-token limit of standard transformer encoders, requiring explicit long-document strategies.
3. **Severe label imbalance.** The label space consists of 115 ICD-10 codes whose frequency ranges from a single training instance to over 1,000.

Evaluation is performed with a *relaxed* micro-F1 metric that groups semantically equivalent ICD-10 codes into clinical clusters, so that a prediction of a code in the same group as the gold label is not fully penalised.

Our system is built around the idea that no single model dominates across all labels and all document types. We therefore combine a neural-only pathway (Greek BERT and XLM-RoBERTa) with symbolic pathways (dictionary rules, information retrieval, named entity recognition) and search for the best label-level fusion strategy using metaheuristic optimisation on the validation set.

The remainder of this paper is structured as follows. Section 2 describes the dataset and preprocessing. Sections 3–7 describe the six individual components in detail. Section 8 presents the ensemble and metaheuristic optimisation. Section 9 reports official and ablation results. Section 10 provides an error analysis, and Section 12 concludes.

2 Dataset and Preprocessing

2.1 Dataset Description

The dataset consists of 2,500 anonymised discharge summaries from Greek cardiology clinics, provided as JSONL files. Each record contains a patient identifier, the free-text discharge summary, and one or more ground-truth ICD-10 codes. The data were split into:

- **Training set:** 2,000 documents
- **Validation set:** 250 documents
- **Test set:** 250 documents
- **Blind test set:** held out by the organisers for official evaluation

The total label space contains 115 ICD-10 codes. The distribution is highly skewed: the five most frequent codes — Z95 (1,135 instances), I25 (1,167), I21 (865), I48 (830), and E11 (712) — account for a disproportionate share of all annotations, while ten codes appear fewer than five times across the entire dataset (E85, I37, I62, I74, M31, M35, R31, R57, T83, Z72). Four codes (E85, I37, R31, R57) appear only once. Table 1 summarises label frequency bands.

Table 1: Label frequency distribution across 115 ICD-10 codes.

Frequency band	# codes	% of total instances
≥ 500 instances	5	$\sim 41\%$
100–499	22	$\sim 36\%$
10–99	58	$\sim 21\%$
< 10 instances	30	$< 2\%$

Documents are moderately long: the mean is 280 tokens, the maximum is 1,218 tokens, and only 2.4% of documents exceed the 512-token limit of standard BERT-style encoders. The average document length is approximately 1,996 characters.

2.2 Stratified Splitting

To ensure label frequency distributions are preserved across splits, we apply multi-label stratified splitting using patient identifiers as the grouping key, preventing records from the same patient from appearing in different splits. Rare labels (fewer than two instances) are handled by forced placement to guarantee at least one training example per class.

2.3 Text Preprocessing

A common preprocessing pipeline is applied to all splits:

1. Unicode normalisation (NFD) and accent stripping
2. Lowercasing (applied only for uncased models such as Greek BERT)
3. Removal of non-alphanumeric characters outside a safe character set (Latin, Greek Unicode blocks, digits, and common punctuation)
4. Whitespace collapse

For XLM-RoBERTa, which uses a case-sensitive SentencePiece tokeniser, only whitespace normalisation is applied to preserve case signals present in abbreviations such as “PCT”, “TAVI”, and “CABG”.

2.4 Data Augmentation

For the XLM-RoBERTa base track, synonym-based augmentation is applied to mitigate class imbalance. Terms in the dictionary are swapped with synonyms at a probability of 0.35. The augmentation multiplier is frequency-adaptive:

Table 2: Frequency-adaptive augmentation multipliers.

Label frequency band	Augmentation factor
≥ 300	$\times 1$ (none)
200–299	$\times 2$
100–199	$\times 3$
30–99	$\times 4$
< 30	$\times 1$ (too few to augment reliably)

Synthetic records are assigned distinct patient identifiers to prevent leakage.

3 Greek BERT Multi-Label Classifier

3.1 Architecture

We fine-tune `nlpaueb/bert-base-greek-uncased-v1`, a BERT-base model pre-trained on a large Greek corpus [1]. The classification head consists of:

1. **Mean pooling** over all token representations (masked by the attention mask), producing a 768-dimensional document embedding
2. **Dropout** (rate 0.3)
3. **Linear layer** $768 \rightarrow 384$ with GELU activation and dropout (0.3)
4. **Linear layer** $384 \rightarrow 115$ producing per-label logits

The maximum sequence length is set to 256 tokens. The mean-pooling strategy was found to outperform CLS-token extraction on the validation set by approximately 0.8 F1 points, because discharge summaries spread diagnostic information across the full document rather than concentrating it in the first sentence.

3.2 Loss Function: Asymmetric Loss

Standard binary cross-entropy treats positive and negative labels symmetrically, which is problematic under severe label imbalance. We use **Asymmetric Loss** (ASL) [2], which independently controls the focal down-weighting of easy negative and positive examples:

$$\mathcal{L}_{\text{ASL}} = \begin{cases} (1 - p)^{\gamma^+} \log(p) & \text{if } y = 1 \\ (p_m)^{\gamma^-} \log(1 - p_m) & \text{if } y = 0 \end{cases} \quad (1)$$

where $p_m = \max(p - m, 0)$ applies a probability margin shift to discard very easy negatives. We use $\gamma^- = 4$, $\gamma^+ = 1$, and a clip value of 0.05. The asymmetric weighting aggressively down-weights easy negative examples (the dominant majority) while applying milder focusing on positive examples, resulting in improved recall on rare labels compared to standard focal loss. The positive weight cap is set to 8.0 to avoid excessive gradient magnitudes for the rarest labels.

3.3 Training Configuration

Table 3: Greek BERT training hyperparameters.

Hyperparameter	Value
Base model	nlpaueb/bert-base-greek-uncased-v1
Max sequence length	256 tokens
Epochs	30 (early stopping patience 7)
Batch size	16
Gradient accumulation	2 steps (effective batch: 32)
Learning rate	3.0×10^{-5}
LR decay (LLRD)	factor 0.9 per encoder layer
Warmup	6% of total training steps
Weight decay	0.02
Max gradient norm	1.0
Dropout	0.3
Precision	FP16 mixed precision
Seed	42

Layer-wise learning rate decay (LLRD). The learning rate is not uniform across layers. Starting from the classification head, each encoder layer is multiplied by a factor of 0.9, so the effective LR for encoder layer l (counting from the output) is $\text{LR} \times 0.9^l$. This ensures that lower layers, which encode general Greek language structure learned during pre-training, are updated more conservatively than the upper task-specific layers. On this dataset, LLRD provided a consistent improvement of roughly 0.4 F1 points on validation compared to uniform learning rate.

3.4 Threshold Tuning

After training, per-label decision thresholds are tuned on the validation set in two passes:

1. **Global sweep:** a single threshold $t \in [0.45, 0.75]$ (step 0.01) is applied to all labels; the value maximising validation micro-F1 is selected as the starting point.
2. **Per-label sweep:** for each label independently, the threshold is swept in the same range. Labels with fewer than 15 positive validation instances are not tuned individually (the global threshold is retained) to avoid overfitting to insufficient statistics. A label-level gain of at least 0.001 F1 is required before accepting a per-label value.

The resulting per-label thresholds range from 0.54 to 0.75, reflecting the model’s varying confidence calibration across labels. Frequent labels (e.g. I21) converge to lower thresholds around 0.54, while rare labels settle near the upper bound of 0.75, consistent with the model’s tendency to under-predict rare codes.

The Greek BERT model achieves a best validation micro-F1 of approximately **0.8165** after per-label threshold tuning.

3.5 Training Variants

Several variants of the Greek BERT model were trained across development phases:

- **p2_long_asl_warmlr**: Early variant with longer max length and warm LR schedule, used as a baseline.
- **p4_asl_mid_reg**: Mid-regularisation variant with weight decay 0.01.
- **p4_winner_lean_head (final)**: Mean pooling, MLP head 768 $\rightarrow$ 384, dropout 0.3, weight decay 0.02. Best overall.
- **mlc_greek_bert_100**: The winner configuration trained for 100 epochs without early stopping, submitted as a standalone system.

4 XLM-RoBERTa Multi-Label Classifiers

4.1 Motivation

While Greek BERT is purpose-built for Greek, its 768-dimensional hidden size and uncased tokenisation may limit capacity on long documents with mixed Greek–Latin terminology. XLM-RoBERTa [3], trained on 100 languages with a 250K SentencePiece vocabulary, natively handles code-switching between Greek medical text and abbreviations such as “PCI”, “TAVI”, and “ACS”. We include both the base (125M parameters) and large (560M parameters) variants.

4.2 Long-Document Handling: Sliding Window

For documents exceeding 512 tokens, we apply a **sliding window** with stride 256. Each chunk shares 256 tokens of overlap with its adjacent chunk, ensuring continuity of context across boundaries. During training, one chunk is selected randomly per document (encouraging the model to produce reliable predictions from any position). During inference, all chunks are processed and their logits are fused:

$$\hat{y}_{\text{fused}} = \alpha \cdot \bar{y}_{\text{mean}} + (1 - \alpha) \cdot y_{\text{max}} \quad (2)$$

with $\alpha = 0.5$ (equal mean–max fusion). The mean term captures the overall label distribution across the document, while the max term ensures that strong local signals from any chunk (e.g. a rare diagnosis mentioned only once) are not diluted by the averaging. Since only 2.4% of documents exceed 512 tokens, the sliding window primarily acts as a safety net rather than a core design requirement.

4.3 XLM-RoBERTa Large

The large variant uses the **ZLPR** loss (Zero-Logit Paired Ranking), a focal-style ranking loss that encourages larger margin between positive and negative label scores without requiring calibrated probability estimates. This is particularly beneficial for multi-label settings with many possible negative classes per document.

Table 4: XLM-RoBERTa Large training hyperparameters.

Hyperparameter	Value
Base model	xlm-roberta-large
Max sequence length	512 tokens
Sliding window stride	256 tokens
Chunk fusion	mean-max ($\alpha = 0.5$)
Epochs	40 (early stopping patience 5)
Batch size	2
Gradient accumulation	16 steps (effective batch: 32)
Learning rate	2.0×10^{-5}
Warmup	6% of total training steps
Weight decay	0.05
Classifier dropout	0.15
Loss	ZLPR
Precision	FP16 mixed precision

The small physical batch size of 2 with 16-step gradient accumulation was necessary due to the large model size (560M parameters). The higher weight decay (0.05) compared to Greek BERT compensates for the larger model capacity and helps prevent overfitting on the small training set.

4.4 XLM-RoBERTa Base

The base variant follows the same sliding-window architecture with a lighter configuration: batch size 16, standard Asymmetric Loss, and optional K-fold cross-validation for generating fold-level predictions. It serves as a computationally cheaper alternative to the large variant and contributes diversity to the ensemble.

5 Information Retrieval Module

5.1 Motivation and Overview

Retrieval-based methods approach coding as a *ranking problem*: given a discharge summary, retrieve the ICD-10 codes whose textual descriptions are most similar to the query. Unlike supervised classifiers, retrieval requires no labelled training data and generalises naturally to rare and unseen codes. Its weakness is lower precision on common codes where distributional patterns are more diagnostic than surface similarity.

The IR module combines three retrieval channels fused via Reciprocal Rank Fusion (RRF).

5.2 BM25 Retrieval

BM25 (Okapi BM25) is a probabilistic sparse retrieval model that ranks ICD-10 code descriptions by term overlap with the query document. A custom tokeniser shared with the dictionary module normalises both query and code descriptions. BM25 scores are computed per code and the top candidates are passed to the fusion stage.

5.3 TF-IDF Retrieval

A TF-IDF vectoriser (scikit-learn) is built over the ICD-10 code description corpus. Cosine similarity between the query TF-IDF vector and each code description vector is computed, and the top candidates are ranked. The TF-IDF channel complements BM25 by assigning term importance weights globally (across the entire code description corpus) rather than locally within the query.

5.4 Dense Retrieval

A multilingual sentence-transformer model (`paraphrase-multilingual-MiniLM-L12-v2`) encodes both the query and code descriptions into 384-dimensional L2-normalised embeddings. Cosine similarity is used for ranking. The dense channel captures semantic similarity that is invisible to sparse models, including synonymous terminology and paraphrases common in clinical language.

5.5 Reciprocal Rank Fusion

The three ranked lists are fused using Reciprocal Rank Fusion (RRF):

$$\text{RRF}(d) = \sum_i \frac{w_i}{k + \text{rank}_i(d)} \quad (3)$$

with $k = 60$ (standard RRF constant), equal weights $w_i = 1.0$ for all channels, and a per-channel candidate pool of size $\max(6 \cdot \text{top_k}, 80)$. The final prediction is obtained by thresholding the RRF score: codes exceeding a minimum fraction of the top score (`fraction_of_top_score=0.22`) are selected, up to a maximum of `max_codes=12` predictions per document.

5.6 Results

The IR module achieves a validation micro-F1 of approximately 0.45 after tuning. As a standalone system its precision is limited (0.31) but its recall is high (0.83), making it a useful diversity contributor in the ensemble for increasing coverage on rare codes. Table 5 shows the effect of the two key tuning parameters.

Table 5: IR module tuning results on the validation set.

Configuration	max_codes	Recall	Precision	F1
Plain top-25	25	0.374	0.086	0.140
Tuned hybrid	12	0.828	0.310	0.451

The large gap between plain and tuned configurations demonstrates that the number of codes retrieved per document is the dominant factor: retrieving 25 codes yields high recall but catastrophic precision, while restricting to 12 codes with a relative score threshold achieves a much better precision-recall balance.

6 Dictionary Baseline

6.1 Term–Code Mapping

The dictionary module performs rule-based coding based on a curated term-to-code mapping stored in `full_dictionary.csv` (1,680 unique term-to-code pairs). A second file, `full_dictionary.train_only.csv`, contains terms derived exclusively from training annotation patterns, used to prevent label leakage when evaluating on validation and test sets.

At inference, an Aho–Corasick automaton performs efficient multi-keyword substring matching over the discharge summary text. Word-boundary checking is enforced to prevent fragment matches. A blacklist of 35 common false-positive terms is applied.

6.2 Rule-Based Code Boosters

In addition to dictionary matching, a set of cardiology-specific pattern rules adds codes based on procedure or diagnosis mentions. Table 6 lists the most impactful boosters.

Table 6: Selected rule-based code boosters (trigger patterns translated from Greek).

Trigger pattern	Added ICD-10 codes
pci, ppci, ptca, angioplasty	Y84
stent, pacemaker, ppm	Z95
tavi	Y84, Z95, I35
cabg	Y84, Z95, I25
myocarditis	I41
pericarditis	I30
pulmonary embolism	I26
ventricular tachycardia	I47
mobitz ii, atrioventricular block	I44
acs, acute coronary syndrome	I20, I21, I22, I24
acute renal failure	N17
haemodialysis, artificial kidney	Z99
respiratory failure	J96
amyloidosis	E85

6.3 Co-occurrence Rules

Certain ICD-10 codes are medically related and frequently co-occur. We add a small set of co-occurrence rules:

- If I10 or I11 is predicted, add the other (hypertensive disease hierarchy)
- If I22 is predicted, add I21 (re-infarction implies prior infarction)
- If I25 is predicted, add Z95 (chronic ischaemic disease often involves implants)

6.4 Optional Filters

The module supports optional post-processing filters that are disabled in the primary configuration but available for ablation:

- **Family history section filter:** suppresses codes matched in sections identified as family history, since those codes apply to relatives rather than the patient.
- **Negation filter:** removes predictions preceded within a 4-token window by negation triggers such as “without”, “negative”, “absent”, “none” (Greek equivalents).
- **Fuzzy matching:** optional Levenshtein-based matching at similarity threshold 85 for handling spelling variation and OCR noise.

6.5 Results

On the training set, the dictionary module achieves a micro-F1 of **0.7176** (precision 0.6534, recall 0.7956), covering 99.76% of documents and 73.04% (84 out of 115) of label types. The 31 uncovered label types include codes with no distinctive lexical surface forms in cardiology text (e.g. metabolic, haematological, and orthopaedic codes). Table 7 summarises the detailed statistics.

Table 7: Dictionary baseline detailed results on the training set.

Metric	Value
Micro-F1	0.7176
Precision	0.6534
Recall	0.7956
True positives	11,415
False positives	6,054
False negatives	2,932
Document coverage	99.76% (2,494/2,500)
Code type coverage	73.04% (84/115 codes)

7 Named Entity Recognition and Entity Linking

7.1 Architecture

The NER+EL pipeline decomposes the coding task into two sub-tasks: (1) identify entity *mentions* in text (NER), and (2) link each mention to an ICD-10 code (EL). This two-stage approach allows modular development and provides interpretable intermediate outputs (explicit mention spans).

NER Component. We fine-tune `nlpaueb/bert-base-greek-uncased-v1` as a sequence tagger using BIO encoding. A partial CRF layer enforces label consistency constraints across adjacent tokens. Training uses 6 epochs, batch size 8, learning rate 2.0×10^{-5} , weight decay 0.01, and early stopping with patience 2. Dynamic padding is applied to reduce computation on shorter sequences.

Entity Linker. For each recognised mention, candidate ICD-10 codes are retrieved via three sources ranked and fused:

1. **Prior-based scoring:** a frequency counter of (mention text, ICD-10 code) pairs built from training annotations. This provides a strong and computationally cheap prior for common cardiology terms.
2. **Dictionary candidates:** the Aho–Corasick matcher from the dictionary module expands mentions to candidate codes via direct term matching.
3. **Context reranker:** a 200-character window surrounding each mention is encoded by `paraphrase-multilingual-MiniLM-L12-v2`. The cosine similarity between the context embedding and the embedding of each candidate code description is computed and combined with the prior score:

$$\text{score}(m, c) = \alpha \cdot \text{prior}(m, c) + (1 - \alpha) \cdot \cos(\mathbf{e}_m, \mathbf{e}_c) \quad (4)$$

with $\alpha = 0.6$ (60% prior, 40% semantic similarity).

7.2 Document-Level Aggregation

Entity-level predictions are aggregated to the document level: a code is predicted for a document if at least one mention was linked to it above a minimum confidence threshold. An optional document-level boost further promotes codes that are also strong dictionary matches in the same document, combining the NER-level and document-level evidence.

7.3 Results

The NER+EL pipeline achieves a best validation micro-F1 of **0.7223** at training epoch 5. It performs comparably to the dictionary baseline but brings complementary evidence through mention-level context, which is particularly useful for distinguishing overlapping ICD-10 categories (e.g. I21 vs. I22 vs. I25).

8 Metaheuristic Ensemble

8.1 Motivation

The six components described in Sections 3–7 have complementary strengths: Greek BERT and XLM-RoBERTa dominate on frequent labels, the dictionary module covers procedural codes with high precision, IR provides broad recall, and NER+EL excels at mention-level disambiguation. No single component is uniformly best, suggesting that ensemble fusion can improve over any individual system.

8.2 Ensemble Architecture

The ensemble module receives six binary prediction matrices (one per component, shape $N_{\text{docs}} \times 115$) and optionally soft scores from the neural components (which provide calibrated probabilities). It outputs a single fused prediction matrix. The full strategy space is defined declaratively in YAML, allowing new strategies to be evaluated without code changes.

The six committee members are:

1. XLM-RoBERTa Large
2. Greek BERT (lean head, per-label thresholds)
3. XLM-RoBERTa Base
4. Information Retrieval (hybrid RRF)
5. Dictionary Baseline
6. NER + Entity Linking

8.3 Base Strategies

We implement ten base fusion strategies:

1. **Weighted ensemble.** A weighted sum of soft scores (or binary votes) is thresholded globally. Weights are optimised on the validation set via metaheuristic search.
2. **Weighted with majority restarts.** Multiple random initialisations of the weighted ensemble are run; the best-performing result across restarts is retained.
3. **Per-label champion routing.** For each label independently, the component with the highest per-class validation F1 is selected. This avoids the averaging effect of global weighting for labels where one model is strongly dominant.
4. **Per-label champion + vote.** The champion prediction is combined with a threshold-filtered majority vote from the remaining components.
5. **Per-patient score routing.** For each document, the component with the highest mean confidence score is used. This adapts to document-level characteristics: some documents are more “retrieval-friendly” while others benefit more from neural classifiers.
6. **Gated secondary (IR/NER).** Primary neural models (Greek BERT, XLM-RoBERTa large) handle high-confidence documents; IR and NER+EL are activated as secondary sources for uncertain cases.
7. **Frequency-bucketed thresholding.** Labels are grouped by training frequency (rare, medium, frequent); a separate threshold is tuned for each bucket.
8. **Dual-threshold.** A high-confidence zone (predict), a low-confidence zone (abstain), and a middle band (require secondary evidence).
9. **Top-K cutoff.** Only the top K scoring labels per document are predicted. K is tuned on the validation set.
10. **Label correction.** A correction grid search maps frequent (predicted, gold) pairs to systematic adjustments, capturing label co-occurrence patterns not modelled by any individual component.

8.4 Composition Operators

Individual strategies can be composed using label-set algebra:

- **Union (merge_or):** Predict a label if *any* constituent strategy predicts it. Maximises recall.
- **Intersection (merge_and):** Predict a label only if *all* constituent strategies agree. Maximises precision.
- **k -of- n voting (merge_k2):** Predict a label if at least k out of n strategies predict it. Balances precision and recall.

The four composed strategies submitted to the shared task are:

1. **merge_or_per_label_weighted:** per-label routing $\cup$ weighted ensemble
2. **merge_or_per_label_vote_weighted:** per-label champion+vote $\cup$ weighted
3. **merge_and_weighted_correction:** weighted $\cap$ correction (best overall)
4. **merge_k2_weighted_per_label_correction:** 2-of-3 vote among weighted, per-label routing, and correction

8.5 Metaheuristic Weight Optimisation

For the weighted ensemble and its variants, component weights cannot be set by hand due to the high-dimensional interaction space. We use a two-stage metaheuristic:

Stage 1: Random search with hill-climbing. Weights are sampled uniformly from the probability simplex ($\sum_i w_i = 1, w_i \geq 0$). Each sampled solution is locally refined by hill-climbing: a small perturbation is applied to each weight pair (one increases, one decreases) and accepted if validation micro-F1 improves. The procedure runs for 10,000 total evaluations with 2 random restarts.

Stage 2: Variable Neighbourhood Search (VNS). Starting from the hill-climbing solution, VNS iterates over a sequence of neighbourhood structures of increasing size:

1. **Shake:** a random perturbation proportional to the neighbourhood index is applied to the current weight vector.
2. **Local search:** hill-climbing is applied within the perturbed neighbourhood.
3. **Accept:** if the local optimum improves the validation micro-F1, move to it and reset the neighbourhood index; otherwise increment the index.

VNS is particularly effective here because the validation micro-F1 landscape has many local optima. The shaking step allows escape from poor local optima without the full cost of random restarts.

Per-cluster routing (optional extension). Documents are clustered by text embedding similarity (K-means with $K \in \{2, 6, 10, \dots, 64\}$) and a separate ensemble weight vector is assigned per cluster. This allows the ensemble to behave differently for post-operative summaries versus primary diagnostic summaries, for example.

8.6 Strategy Selection and Submission

All 14 strategies (10 base + 4 composed) are evaluated on the validation set. A manifest file records the validation score, component weights, per-label thresholds, and any errors for each strategy. The five strategies with the highest validation micro-F1 are selected as official submissions.

9 Experiments and Results

9.1 Individual Component Performance

Table 8 summarises the performance of each component on the validation set before any ensembling.

Table 8: Individual component validation micro-F1.

Component	Val. micro-F1	Notes
Greek BERT (lean head + per-label thr.)	0.8165	Primary neural model
NER + Entity Linking	0.7223	Best at epoch 5
Dictionary Baseline	0.7176	99.76% document coverage
Information Retrieval (hybrid RRF)	0.4510	Recall 0.83, Precision 0.31

The Greek BERT model is the strongest individual component by a large margin. The dictionary and NER+EL modules perform comparably and provide strong symbolic evidence. The IR module has the lowest standalone F1 but contributes high recall that helps the ensemble recover missed labels.

9.2 Ablation: Effect of Threshold Tuning

Table 9 isolates the contribution of threshold tuning for Greek BERT. Moving from a fixed threshold to a global sweep yields roughly one F1 point, and the additional per-label sweep adds another 0.5–1.0 points.

Table 9: Threshold tuning ablation for Greek BERT (validation micro-F1).

Threshold strategy	Validation micro-F1
Fixed $t = 0.5$	≈ 0.79
Global sweep $[0.45, 0.75]$	≈ 0.80
Per-label sweep, 2-pass	0.8165

9.3 Ablation: Effect of LLRD

Layer-wise learning rate decay consistently improved Greek BERT validation micro-F1 by approximately 0.4 F1 points compared to a uniform learning rate. This improvement was more pronounced for rare labels, where the lower encoder layers’ general language representations are more load-bearing.

9.4 Ablation: Loss Function Comparison

We compared Asymmetric Loss (ASL) against binary cross-entropy (BCE) and focal loss on the Greek BERT model. ASL outperformed both alternatives on the validation set, with the improvement primarily driven by better recall on rare labels (those appearing fewer than 50 times in training). BCE showed good precision but lower recall; focal loss improved recall compared to BCE but less so than ASL.

9.5 Ablation: Ensemble Strategy Comparison

Table 10 compares the ensemble composition operators. The intersection strategy (`merge_and`) achieves the highest precision and the best overall F1, at the cost of slightly reduced recall compared to union-based strategies.

Table 10: Ensemble composition operator comparison (schematic on validation set).

Strategy	Recall	Precision	F1 trend
Union (<code>merge_or</code>)	highest	lowest	moderate
Weighted ensemble	mid	mid	strong
k -of- n (<code>merge_k2</code>)	mid	mid+	strong
Intersection (<code>merge_and</code>)	mid-	highest	best

The intersection of the weighted ensemble with the label correction module (`merge_and_weighted_corr`) is the best overall strategy: the weighted ensemble provides strong baseline predictions, and the correction module further filters systematic false positives, boosting precision without hurting recall.

9.6 Official Test Results

Table 11 reports all our submissions on the official ElCardioCC 2026 test set.

Table 11: Official ElCardioCC 2026 test set results.

Submission	Recall	Precision	F1
ensemble (safer thr + <code>merge_and</code> weighted corr.)	0.8510	0.8830	0.8667
ensemble (extended, <code>merge_k2</code> weighted per-label corr.)	0.8687	0.8539	0.8613
ensemble (safer thr, <code>merge_k2</code> weighted per-label corr.)	0.8767	0.8431	0.8596
mlc_greek_bert_100 (100 epochs)	0.8194	0.8811	0.8491
mlc_greek_bert (30 epochs)	0.8310	0.8675	0.8489

The ensemble gain over the single best Greek BERT model is approximately **+1.8 F1 points** (0.8667 vs. 0.8489). The two standalone Greek BERT submissions achieve nearly identical F1 despite the 100-epoch variant having longer training, suggesting diminishing returns beyond convergence and no significant overfitting from extended training.

Comparing the three ensemble submissions, the precision-oriented `merge_and` strategy achieves the highest F1, while the `merge_k2` variants achieve higher recall at the cost

of precision. The precision advantage of `merge_and` is consistent with its design intent: requiring agreement between the weighted ensemble and the correction module filters out predictions where the two strategies disagree, typically false positives.

10 Error Analysis

10.1 Rare Label Performance

The 30 labels with fewer than 10 training instances are the primary source of errors. For 10 of these (E85, I37, I62, I74, M31, M35, R31, R57, T83, Z72), fewer than 15 positive validation instances are available, making per-label threshold tuning unreliable. The model tends to either miss these labels entirely (recall ≈ 0) or over-predict them, contributing disproportionately to precision losses.

The dictionary module partially compensates: the explicit rule linking the term “amyloidosis” to E85 recovers the single training instance of E85 with high precision. However, for codes with no distinctive lexical surface forms (e.g. M35, R57), no component achieves reliable performance.

10.2 Dictionary Coverage Gaps

The dictionary module fails to cover 31 of the 115 label types, including metabolic (E00), haematological (D47), musculoskeletal (M25, M32, M96), and nephrology codes (N02, N93). These codes typically require inference from laboratory values not present in the text, or reasoning over implicit multi-sentence evidence.

10.3 Label Confusion

The most frequent error type is confusion between closely related ICD-10 codes that describe different severity levels or temporal stages of the same condition:

- **I21 / I22 / I25:** Acute myocardial infarction, subsequent MI, and chronic ischaemic disease may all appear in the same document. Accurate assignment requires temporal reasoning (“prior infarction” vs. “current event”) that is not well captured by bag-of-token models.
- **I48 subtypes:** Atrial fibrillation subtypes may be described inconsistently in the text, causing both missed and spurious predictions at the sub-type level.
- **Z95 over-prediction:** This code (presence of cardiac/vascular implants) is heavily over-predicted by the dictionary and IR modules due to pervasive mentions of stents and pacemakers. It is the main source of ensemble imprecision and motivates the `merge_and` correction step.

10.4 Model Confidence Calibration

The Greek BERT model shows systematic under-confidence on rare labels: predicted probabilities for true positive rare-label documents often fall below 0.5. Per-label threshold tuning mitigates this by setting lower decision boundaries, but is unreliable for labels

with fewer than 15 positive validation instances. Future work should explore temperature scaling or isotonic regression calibration specifically for the tail of the label distribution.

11 Training Infrastructure

All experiments were conducted using PyTorch 2.0+ and Hugging Face Transformers 4.36+. FP16 mixed-precision training was enabled for all neural models to reduce memory consumption and accelerate training on GPU. DataLoader workers were set to 4 with persistent workers and pinned memory to maximise GPU utilisation during data loading.

Experiment tracking. All training runs, validation curves, hyperparameter configurations, and model artefacts were logged with **Weights & Biases** under the project `elcardiocc-2026`. The Greek BERT winner run is identified as `greek-bert-p4-winner-lean-head` with tags `{mlc, greek-bert, phase4, lean-head, mean-pool, head-384, dropout-03, wd-002}`.

Artefact organisation. Model checkpoints and best-threshold JSON files are saved under `outputs/models/{model_name}/`. Predictions for all splits (train, validation, test, blind test) are stored under `outputs/predictions/{model_name}/` in JSONL format. Ensemble outputs are organised by strategy slug under `outputs/predictions/ensemble_metaheuristic`.

12 Conclusion

We described AIDA-ElCardio, a multi-component system for automatic ICD-10 coding of Greek cardiology discharge summaries, developed as part of the MSc in Artificial Intelligence and Data Analytics at the University of Macedonia.

The system combines fine-tuned language models (Greek BERT and XLM-RoBERTa) with symbolic components (dictionary rules, hybrid information retrieval, and NER-based entity linking) under a declarative metaheuristic ensemble framework. Individual components address complementary aspects of the task: neural models capture distributional patterns for frequent labels, the dictionary achieves high-precision coverage of procedural codes, and the IR module provides broad recall for rare codes.

The ensemble is optimised via random search with hill-climbing and Variable Neighbourhood Search over a strategy space of ten base strategies and four composition operators. Our best submission achieves a micro-F1 of **0.8667** on the official test set, representing an improvement of approximately 1.8 F1 points over the single best standalone Greek BERT model (0.8489).

The error analysis highlights three main limitations: rare-label under-coverage, dictionary gaps for non-cardiology codes, and label confusion between temporally related ICD-10 codes. Future work will address these through (i) calibration techniques for the long tail of the label distribution, (ii) temporal reasoning modules for distinguishing related codes, (iii) larger language models with native Greek pre-training, and (iv) cross-lingual transfer from well-annotated English clinical corpora.

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